

Trainings ▾

General Information

Thermostat Support

- The thermostat support assembly contains a support and two thermostat housings. Each thermostat housing contains two thermostats. The support assembly also provides for connection of the water outlet tubing for the cylinder head, air compressor, aftercooler, and a mounting provision for coolant filters. The low-temperature aftercooler thermostat housing does **not** include a mounting provision for coolant filters.
- A revised thermostat housing support with 6.35 mm [0.25 inch] vent holes located higher in the support for improved venting replaced the previous housing in February of 1992, engine serial number first 33121370. The support housing with 19.05 mm [0.75 inch] holes remained in production.

Thermostat Seal

- Each thermostat housing contains two thermostat seals.
- The thermostat seal for K38 and K50 engines was changed. The revised seal has a sealing element that is thicker. This sealing element prevents fretting and leakage resulting from the higher coolant flow required in K38 and K50 engines. The thin seal is installed on other Cummins® engines. The thick seal **must** be used on K38 and K50 engines.

Heat Shield

- All KT38, KTA38, and KTA50 engines with a center mount aftercooler or air transfer housing **must** have a heat shield installed on the right bank thermostat housing. This heat shield is required to prevent damage to the outlet hose from the heat of the exhaust outlet of the turbocharger. KTA38 and KTA50 engines with outboard aftercoolers do **not** require the heat shield. The thermostat housing mounted design replaced the coolant outlet hose mounted design in February of 1983, engine serial number first 33107562. A revised thermostat housing was also released to accommodate the revised heat shield. The older heat shield **must not** be installed again or failure of the right bank coolant outlet hose will result.

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid the possibility of arcing, remove the negative (-) battery first, and attach the negative (-) battery cable last.

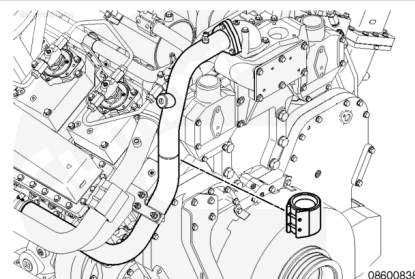
- Disconnect the battery cables. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>).
- Remove the air cleaner assemblies. Refer to Procedure 010-013 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-013-tr.html>)
- Remove the cooling fan, if installed. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-040-tr.html>)
- Remove the fan belt, if installed. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Remove the fan hub, if installed. Refer to Procedure 008-036 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html>)

- Remove the fan hub bracket, if installed. Refer to Procedure 008-035 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html)
- Remove the water filters, if installed. Refer to Procedure 008-006 in Section 8.
(/qs3/pubsys2/xml/en/procedures/102/102-008-006-tr.html)
- Remove the heat exchanger, if equipped. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-053-tr.html)

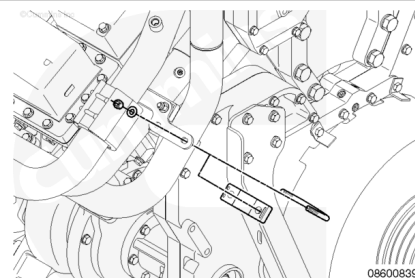
Remove

Low-Temperature Aftercooling Thermostat Support:

Remove the Straub® coupling from the bypass line.



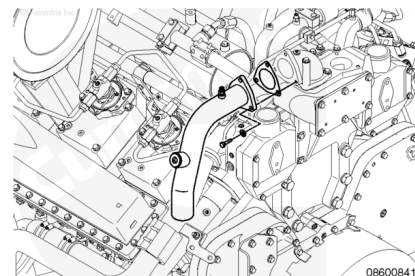
Remove the U-bolts from the bypass tube.



Remove the capscrews and washers from the upper flange connecting the bypass tube to the thermostat housing.

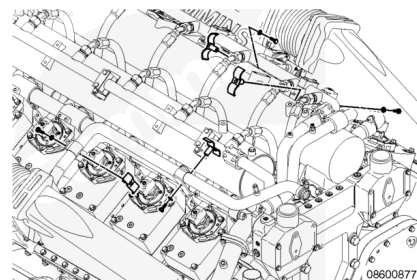
Remove the upper section of the bypass tube.

Discard the gasket.



Note : The following step must be carried out on both engine banks.

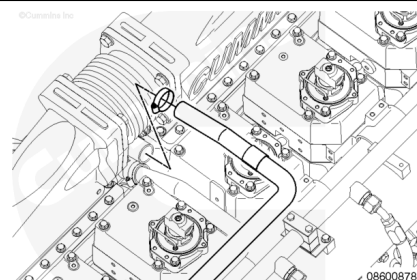
Remove the capscrews and retaining clamps from the water tubes to the aftercoolers.



Note : The following step must be carried out on both engine banks.

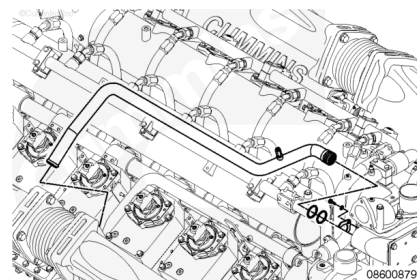
Loosen the hose clamps connecting the transfer tubes to the rubber tee between the aftercoolers.

Separate the transfer tubes from the rubber tees.

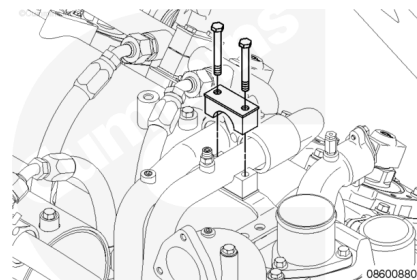


Remove the capscrew and clamp from the right bank upper transfer tube on the LTA thermostat housing.

Remove the transfer tube and discard the o-rings.

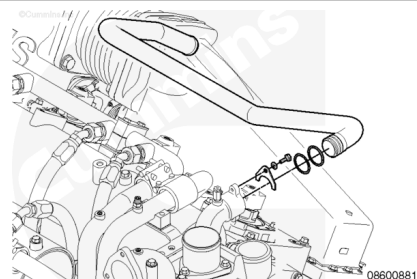


Remove the two capscrews and upper half of the transfer tube isolator.



Remove the one capscrew, washer, and clamp from the left bank transfer tube.

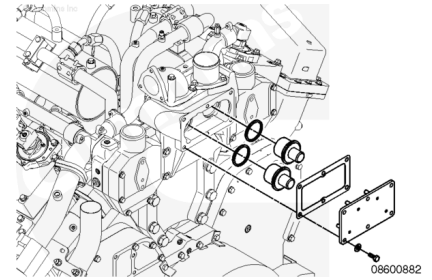
Remove the transfer tube and discard the o-rings.



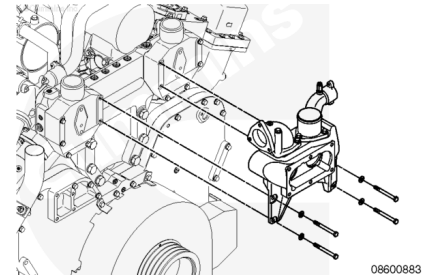
Remove the six capscrews, gasket, and thermostat cover.

Remove the LTA thermostats and seals.

Discard the cover gaskets and thermostat seals.

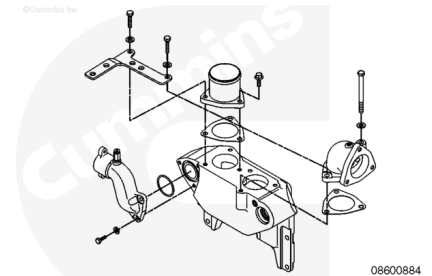


Remove the four mounting capscrews, washers, and LTA thermostat housing.



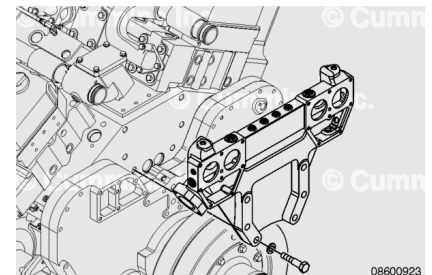
Remove the transfer tube isolation bracket and water outlets from the thermostat housing.

Discard the o-ring and gaskets.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



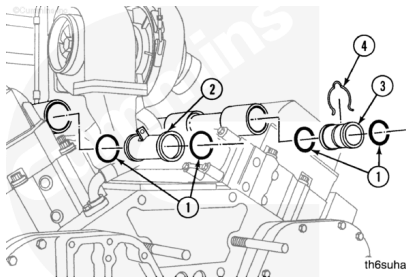
Note : Shown is the K38 and K50 thermostat housing and support bracket. Although different in appearance, the procedure remains the same for the QSK engines.

Remove the six support assembly-to-cylinder block mounting capscrews.

Remove the thermostat support assembly (5).

Remove the water transfer tubes (2) and (3). Remove the o-rings.

Discard the o-rings.

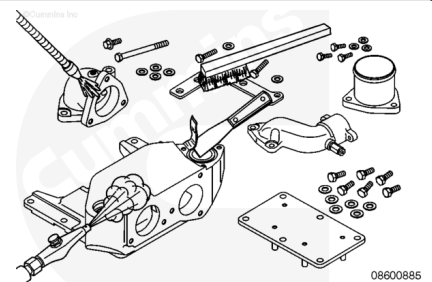


Clean and Inspect for Reuse

LTA

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a suitable scraper to clean all gasket surfaces.

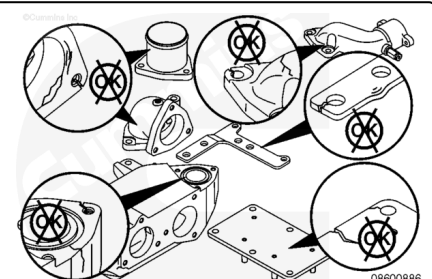
Use a wire brush to clean the LTA thermostat support housing.

Thoroughly clean the o-ring bores.

Dry with compressed air.

Check for cracks or corrosion damage around the o-ring seal bores.

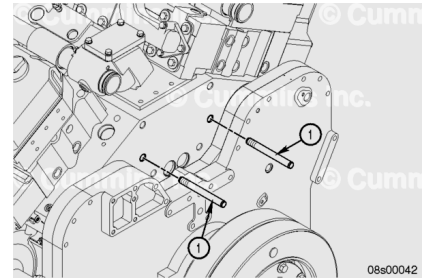
Check for damaged threads and gasket surfaces.



Install

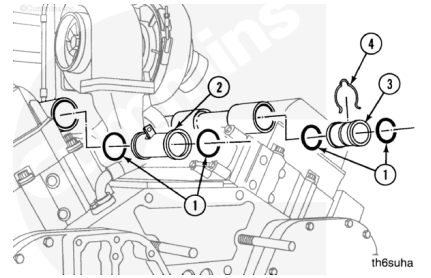
Note : Shown is the K38 and K50 thermostat housing and support bracket. Although different in appearance, the procedure remains the same for the QSK engines.

Install guide pins (1) in two of the six thermostat support assembly holes in the cylinder block for alignment.



⚠ CAUTION ⚠

The retaining clip (4) must be installed on the short water tube (3) for the left bank. If the clip is not installed, the tube can move from the bore and cause a coolant leak, resulting in extensive damage to the engine. The retaining clip is not required for the tube that is installed on the right bank.



Note : If the engine has wet type exhaust manifolds, the tube for the right bank (2) is the same as the short tube for the left bank. The retaining clip (4) is required.

Note : On newer engines with dry exhaust manifolds, the tube for the right bank contains a tab that prevents the tube from passing completely through the bore in the thermostat housing. Be sure that the tube is installed with the tab towards the thermostat support.

Install the two o-rings (1) on the water tube for the right bank (2) and the left bank (3).

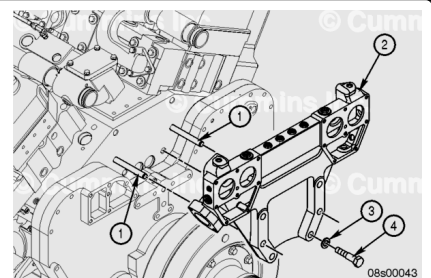
Install the tube clip (4) on the tube for the left bank (3).

Use vegetable oil to lubricate the o-rings.

Install the tubes in the bores of the rocker lever housing.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the thermostat support assembly (2) over the guide pins (1).

Install four lock washers (3) and four capscrews (4) in free holes and secure the assembly to cylinder block.

Tighten the capscrews to specification.

Remove the guide pins and install the remaining two lock washers and two capscrews.

Torque Value: 366 n•m [270 ft-lb]

LTA

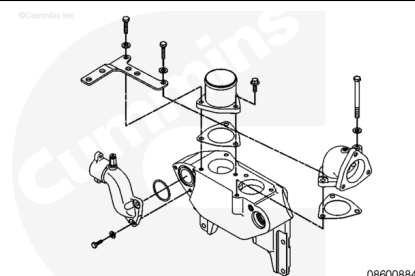
Use new gaskets and o-rings and install the water outlets and transfer tube isolation bracket onto the thermostat housing.

Torque Value:

Water Outlet Capscrews 38 n•m [28 ft-lb]

Torque Value:

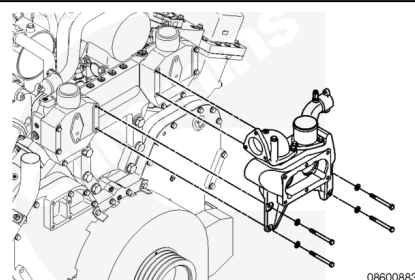
Transfer Tube Capscrews 11 n•m [97 in-lb]



Install the thermostat housing, washers, and capscrews.

Torque Value:

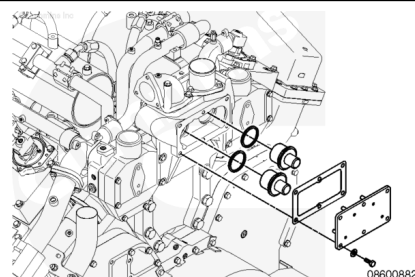
LTA Thermostat Housing Capscrews 38 n•m [28 ft-lb]



Install new thermostat seals, thermostats, new cover gasket, and cover. Install the six washers and capscrews.

Torque Value:

LTA Thermostat Cover Capscrews 60 n•m [44 ft-lb]

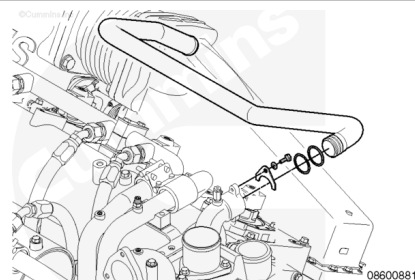


Install two new o-rings on the left bank transfer tube and lubricate them with vegetable oil.

Install the transfer tube, clamp, and one washer and capscrew.

Torque Value:

Left Bank Transfer Tube Capscrews 22 n•m [195 in-lb]

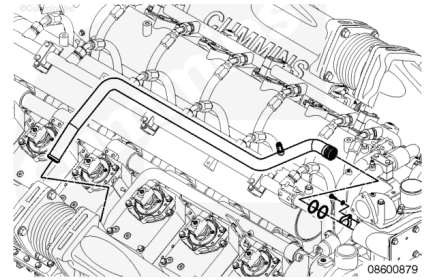


Install two new o-rings on the right bank transfer tube and lubricate them with vegetable oil.

Install the transfer tube, clamp, and one washer and capscrew.

Torque Value:

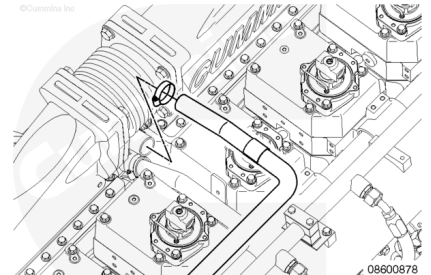
Right Bank Transfer Tube Capscrews 22 n•m [195 in-lb]



Note : The following step must be carried out on both engine banks.

Install the transfer tubes into the rubber tees.

Tighten the hose clamps at the rubber tee between the aftercoolers.



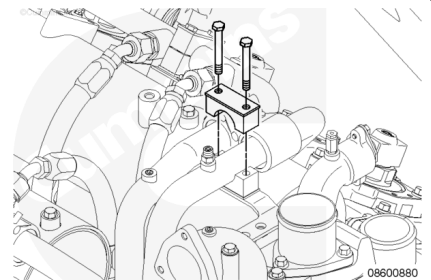
Torque Value:

Hose Clamps 8 n•m [71 in-lb]

Install the upper half of the transfer tube isolator and two capscrews.

Torque Value:

Upper Half Transfer Tube Capscrews 11 n•m [97 in-lb]

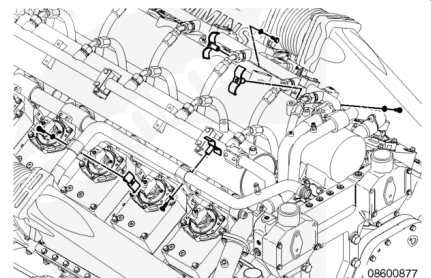


Note : The following step must be carried out on both engine banks.

Install the retaining clamps, washers, and capscrews to the water tubes to the aftercoolers.

Torque Value:

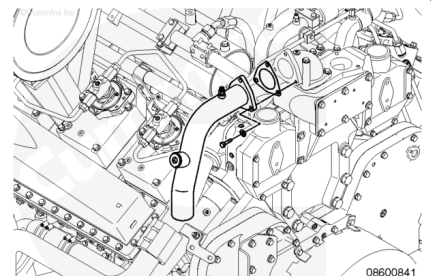
Retaining Clamps 38 n•m [28 ft-lb]



Use a new gasket and install the washers and capscrews on the upper flange connecting the bypass tube to the thermostat housing.

Torque Value:

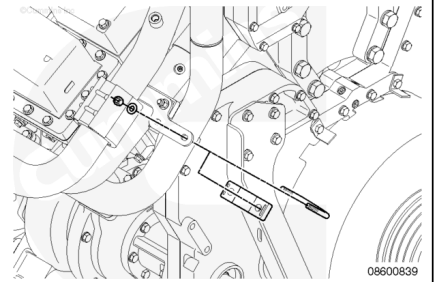
Upper Flange Capscrews 38 n•m [28 ft-lb]



Install the U-bolts on the bypass tube.

Torque Value:

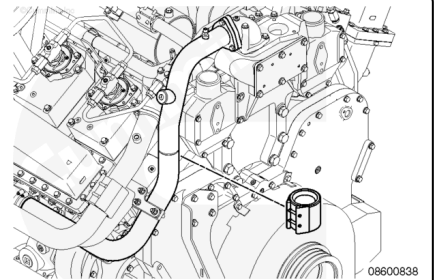
Bypass Tube U-bolts 22 n•m [195 in-lb]



Install the Straub® coupling on the bypass line.

Torque Value:

Straub® Coupling 30 n•m [22 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

**⚠ WARNING ⚠**

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid the possibility of arcing, remove the negative (-) battery first, and attach the negative (-) battery cable last.

- Install the heat exchanger, if equipped. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-053-tr.html)
- Install the water filters, if installed. Refer to Procedure 008-006 in Section 8
(/qs3/pubsys2/xml/en/procedures/102/102-008-006-tr.html).
- Install the fan hub bracket, if installed. Refer to Procedure 008-035 in Section 8

(/qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html).

- Install the fan hub, if installed. Refer to Procedure 008-036 in Section 8 (/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
- Install the fan belt, if installed. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)
- Install the cooling fan, if installed. Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-040-tr.html)
- Install the heat exchanger, if equipped. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-053-tr.html)
- Install the air cleaner assemblies. Refer to Procedure 010-013 in Section 10.
(/qs3/pubsys2/xml/en/procedures/28/28-010-013-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect the battery cables. Refer to the OEM service manual.
- Operate the engine to normal operating temperatures and pressures and check for leaks.